STATISTICAL LEARNING THEORY (03/03/26)

Summer Term 2026

Other course names:

Machine Learning (Data Science Master)
Adv. Machine Learning (AIS Master)

MOTIVATION

Example:

	$\in \mathbb{R}$ Weight (in g)	$\in \mathbb{R}$ Color (in $[0,1]$)	$\in \{0,1\}$ Tasty?
Papaya 1	800	0.1	0
⋮	⋮	⋮	⋮
Papaya N	1200	0.7	1

0 for TASTY? means "not tasty" and 1 means "tasty".

Lets call the information in this table our training data.

Based on that, we try to find

$$\boxed{h: \mathbb{R} \times \mathbb{R} \to \{0, 1\}}$$

i.e., a function that takes a tuple of (weight, color) and outputs a prediction of whether a papaya is tasty (1) or not (0).

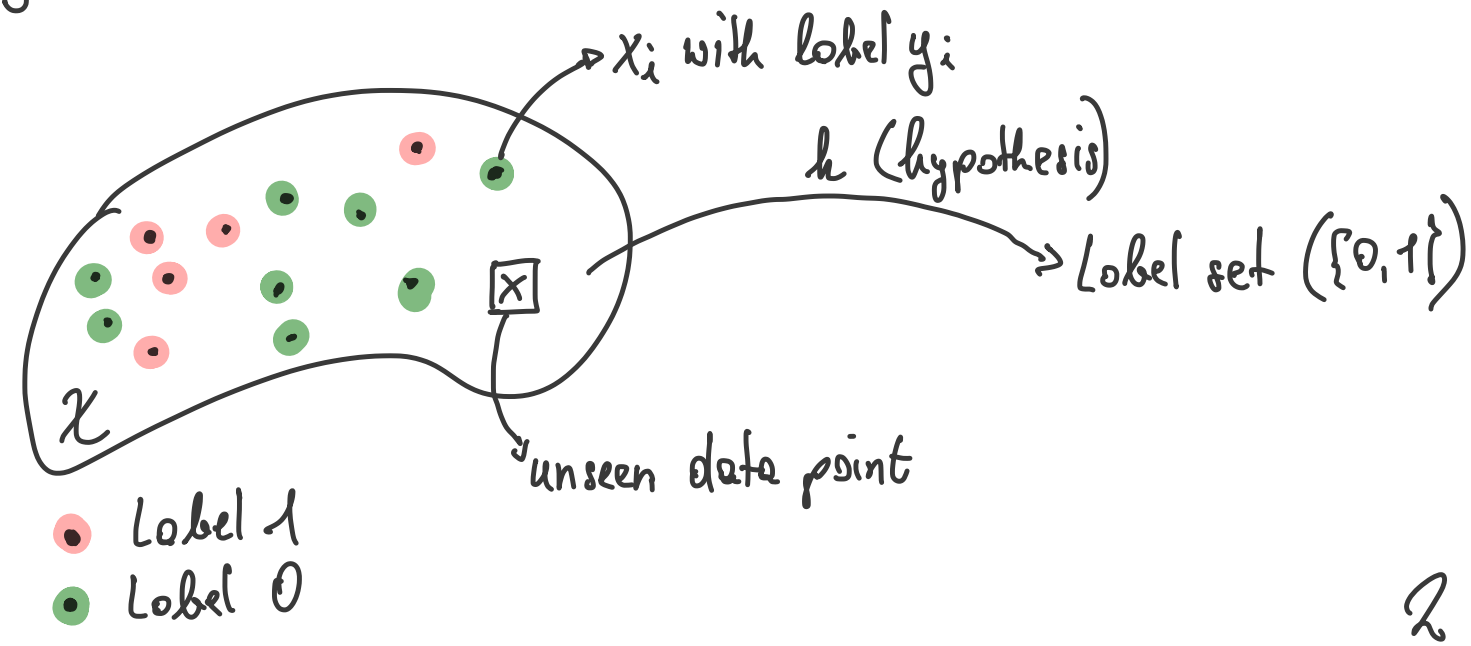
We are going to call such a function (h) a **hypothesis**.

More formally, the available data comes in the form of

$$((x_1, y_1), \ldots\ldots, (x_N, y_N)) = S$$

with (in our example) $x_i \in \underbrace{\mathbb{R}^2}_{\text{domain } \mathcal{X}}$ and $y_i \in \underbrace{\{0, 1\}}_{\text{Label set } \mathcal{Y}}$.

We say a **learner** (some algorithm) receives the training data S and returns h!



Two assumptions we are going to make initially:

A.1. All the x_i's are drawn <u>independently and identically distributed</u> (iid) from some (unknown) distribution D over our domain $\mathcal{X}$.

A.2. The x_i's are labeled by some (unknown) function

$$\boxed{f : \mathcal{X} \to \mathcal{Y}}$$

called the true labeling function.

This means our training data is of the form

$$S = \left((x_1, \underbrace{f(x_1)}_{y_1}), \ldots\ldots, (x_N, \underbrace{f(x_N)}_{y_N}) \right)$$

<u>What do we care about?</u>

true labeling function

$$\{ x \in \mathcal{X} : h(x) \neq f(x) \} = A$$

hypothesis

i.e., all points in our domain $\mathcal{X}$ for which h <u>differs</u> from f.

Specifically, we are interested in the probability of making a mistake $(h(x) \neq f(x))$

$$D(A) = D\left(\{ x \in \mathcal{X} : h(x) \neq f(x) \} \right) = \mathbb{P}_{x \sim D}\left[h(x) \neq f(x) \right]$$

$$(*) \quad \boxed{=: L_{D,f}(h)}$$ generalization error

The <u>empirical</u> version of (*) is

$$\frac{1}{N} \cdot \left| \left\{ i \in \{1, \dots, N\} : h(x_i) \neq f(x_i) \right\} \right| =: \boxed{L_S(h)}$$

We call this the empirical error/risk.

(<u>Notation</u>: $[N] = \{1, \dots, N\}$
$S|_x = (x_1, \dots, x_N)$)

We can also write the empirical error as

$$L_S(h) = \frac{1}{N} \cdot \sum_{i=1}^{N} \mathbb{1}_{h(x_i) \neq f(x_i)} \quad \text{with}$$

$$\mathbb{1}_{h(x) \neq f(x)} = \begin{cases} 1, & \text{if } h(x) \neq f(x) \\ 0, & \text{else} \end{cases}$$

<u>Claim</u> (relating L_S to $L_{D,f}$):

$$\mathop{\mathbb{E}}_{S|_x \sim D^N} \left[L_S(h) \right] = L_{D,f}(h)$$

<u>Proof:</u> next week!

03/10/26

$$\mathbb{E}_{S|x\sim D^m}\left[L_S(h)\right] = \mathbb{E}_{S|x\sim D^m}\left[\frac{1}{N}\cdot\sum_{i=1}^{N}\mathbb{1}_{h(x_i)\neq f(x_i)}\right] \quad \text{// by def.}$$

$$= \frac{1}{N}\cdot\sum_{i=1}^{N}\mathbb{E}_{x_i\sim D}\left[\mathbb{1}_{h(x_i)\neq f(x_i)}\right] \quad \text{// by linearity of } \mathbb{E}[\cdot]$$

$$= \frac{1}{N}\cdot\sum_{i=1}^{N}\mathbb{E}_{x\sim D}\left[\underbrace{\mathbb{1}_{h(x)\neq f(x)}}_{\text{only takes values } 0 \text{ or } 1}\right] \quad \text{// as all } x_i\text{'s are iid}$$

$$= \frac{1}{N}\cdot\sum_{i=1}^{N}\mathbb{P}_{x\sim D}\left[h(x)\neq f(x)\right]$$

$$= \frac{1}{\cancel{N}}\cdot\cancel{N}\cdot\mathbb{P}\left[h(x)\neq f(x)\right]$$

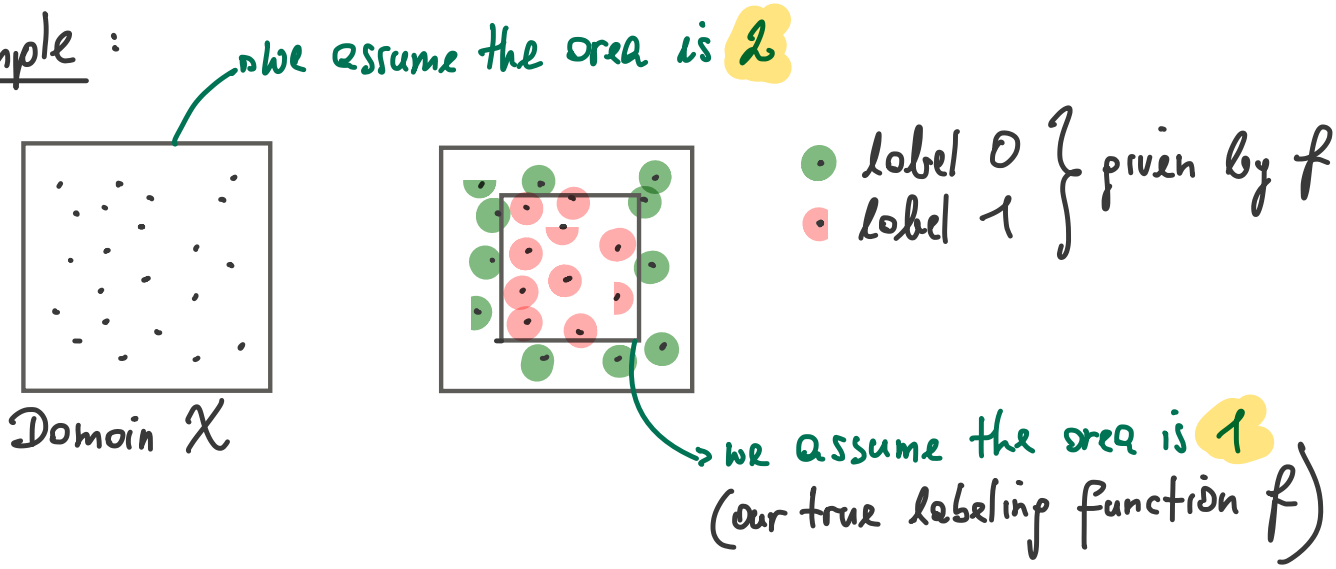
$$= \mathbb{P}_{x\sim D}\left[h(x)\neq f(x)\right]$$

$$= L_{D,f}(h) \quad \text{// by def.}$$

Our first learning paradigm

Empirical risk minimization (ERM): As a learner only has access to S (our training data), it is "natural" to try to select h (our hypothesis) such that the empirical risk/error $L_S(h)$ is minimized. We call such a hypothesis an empirical risk minimizer (h_S)!

Example:



The distribution on $\mathcal{X}$ is a <u>uniform distribution</u>.

Lets say we have an algorithm that returns h_S

$$h_S(x) = \begin{cases} y_i, & \text{if } \exists i \in \{1, \dots, N\} : x_i = x \\ 0, & \text{else} \end{cases}$$

Clearly, h_S is correct on <u>all</u> training data points in $S \Rightarrow h_S$ is an <u>empirical risk minimizer</u>.

$$L_S(h_S) = 0 \,!$$

But, on unseen data points drawn from $\mathcal{D}$ (which is a uniform distribution on $\mathcal{X}$), h_S is only correct 50% of the time $\Rightarrow$

$$L_{\mathcal{D},f}(h_S) = \frac{1}{2} \,!$$

That's what we call overfitting!!!

Hypothesis class ($\mathcal{H}$): We restrict searching for h to $\mathcal{H}$, i.e., a class of functions from $\mathcal{X} \to \mathcal{Y}$ and we write

$$\boxed{ERM_{\mathcal{H}}(S) \in \underset{h \in \mathcal{H}}{\operatorname{argmin}} \; L_S(h)}$$

ERM over finite hypothesis classes ($|\mathcal{H}| < \infty$)

Assumption A.3 : $\exists h^* \in \mathcal{H}$ with $L_{\mathcal{D},f}(h^*) = 0$!
(realizability)

Now, any ERM hypothesis h_S will attain 0 empirical error (meaning $L_S(h_S) = 0$!), as h_S competes against h^* and h^* has $L_S(h^*) = 0$ (and also $L_{\mathcal{D},f}(h^*) = 0$ by A.3).

We know that $L_{\mathcal{D},f}(h_S) > \varepsilon$, $\varepsilon \in (0,1)$, can only happen if our learner selects a hypothesis h_S with $L_S(h_S) = 0$, <u>but</u> $L_{\mathcal{D},f}(h_S) > \varepsilon$.

We define

$$\mathcal{H}_{BAD} = \{ h \in \mathcal{H} : L_{\mathcal{D},f}(h) > \varepsilon \}$$

(set of BAD hypotheses)

We also define

$$M = \{ S|_x : \exists h \in H_{BAD},\ L_S(h) = 0 \}$$

("misleading" training data sets)

<u>Observation:</u>

ERM hypothesis

$$\{ S|_x : L_{D,f}(h_S) > \varepsilon \} \subseteq \{ S|_x : \exists h \in H_{BAD},\ L_S(h) = 0 \} = M$$

Since

$$\{ S|_x : \exists h \in H_{BAD},\ L_S(h) = 0 \} = \bigcup_{h \in H_{BAD}} \{ S|_x : L_S(h) = 0 \}$$

we have

$$D^m\left(\{ S|_x : \exists h \in H_{BAD},\ L_S(h) = 0 \} \right) = D^m\left(\bigcup_{h \in H_{BAD}} \{ S|_x : L_S(h) = 0 \} \right)$$

$$\leq \sum_{h \in H_{BAD}} D^m\left(\{ S|_x : L_S(h) = 0 \} \right)$$

(by the "union bound")

Let us fix one $h \in H_{BAD}$ for a moment!

$$\underbrace{\mathcal{D}^N\left(\{S|_x : L_S(h)=0\}\right)}_{\text{one of the summation terms}} = \mathcal{D}^N\left(\{S|_x : \forall i \in \{1,\dots,N\} : h(x_i)=f(x_i)\}\right)$$

$$= \prod_{i=1}^{N} \mathcal{D}\left(\{x \in \mathcal{X} : h(x) = f(x)\}\right)$$

(by our iid assumption A.1)

Remember that $h \in H_{\text{BAD}}$, so $L_{\mathcal{D},f}(h) > \varepsilon$.

$$= \prod_{i=1}^{N} \left(1 - \underbrace{L_{\mathcal{D},f}(h)}_{>\varepsilon}\right)$$

$$\leq \prod_{i=1}^{N} (1-\varepsilon)$$

$$= (1-\varepsilon)^N$$

$$\leq e^{-\varepsilon N} \quad \text{(next week)}$$